

SHUBHAM R CHOWTA

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EDUCATION

Sahyadri College of Engineering & Management

2023 – 2027

B.Tech in Computer Science & Engineering (AI & ML) | CGPA: 8.77/10

Languages: German (11), Japanese (10), Chinese (10), Italian (9) | 880+ day Duolingo streak

TECHNICAL SKILLS

Languages: C, Python, Java, SQL

CS Fundamentals: Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks

Systems / Backend: Linux, WebAssembly, Syscalls, VFS, Process Scheduling, FastAPI, Flask, REST APIs, SQLite, Git

ML / AI: PyTorch, TensorFlow, TFLite, CNNs, Reinforcement Learning, RAG

SOFTWARE ENGINEERING EXPERIENCE

Software Engineering Training Program

2024 – 2026

Sahyadri College of Engineering & Management

- Selected as **1 of 60** students from a cohort of 600 for a multi-year program focused on real-world software engineering and team-based development.
- Served as Scrum Master for a 5-member team while contributing to a banking application using Python, FastAPI, SQL, SQLite, REST APIs, and unit testing.
- Worked across software testing, data processing, data engineering, KNIME, and LangChain through iterative projects, reviews, and technical presentations.

PROJECTS

SA2 — Heterogeneous Distributed Deep Learning Framework

Jul 2025 – Present

Python, PyTorch, FastAPI, Soft Actor-Critic

- Led a team of 4 building a distributed deep-learning framework for heterogeneous high-end and resource-constrained devices.
- Designed a time-bounded, staleness-aware local-training protocol so slower devices contribute without blocking global synchronization; handled **300+** thermal-throttling and straggler events per session.
- Engineered a Soft Actor-Critic scheduler achieving **94.07% MNIST test accuracy** within a **5-minute training budget** across **3 laptops and 3 phones**, versus **84.62%** for synchronous FedAvg (**+9.45 percentage points**).

RuneOS — Browser-Based Experimental Operating System

Jul 2024 – Present

C, WebAssembly, JavaScript

- Implemented a monolithic C kernel with syscall interfaces, process lifecycle management, and a cooperative round-robin scheduler.
- Built a tree-based VFS supporting directories, files, and pseudo-filesystems including `/proc` and `/kernel`; compiled the kernel to WebAssembly for browser execution.

Arecanut Disease Classification — Edge ML Deployment

Jan 2025 – Jul 2025

Python, TensorFlow, TFLite, React Native

- Led a team of 4 developing a CNN for real-world arecanut disease classification, achieving **98.5%** accuracy; quantized and deployed it to TFLite for offline React Native inference.
- Evaluated mobile-camera to UAV-imagery domain shift, achieving **91.3%** accuracy and analyzing quantization and preprocessing effects.

GEM — Agentic Personal Assistant

Oct 2024 – Present

Python, LlamaIndex, Prompt Engineering

- Initially developed with a team of 6 using LlamaIndex; subsequently **rebuilt solo** around Markdown-based memory, retrieval, embeddings, and custom tool execution.
- Designed the agentic execution workflow independently rather than relying on a prebuilt agent framework, and deployed the system for internal engineering-club use.

COMMUNITY & LEADERSHIP

Team Challengers: Member of a multidisciplinary engineering club; collaborated across technical interests and contributed to organizing technical events.